Altermagnets and spin liquids

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Abstract

What happens to an altermagnet when quantum fluctuations destroy its spin order? This report addresses this question following the work of Sobral, Mandal, and Scheurer [1]. Starting from a checkerboard J_1-J_2-K Heisenberg model whose two sublattices are related by a fourfold rotation instead of a translation, we first map out the classical phase diagram. Besides collinear and nematic phases it contains a noncoplanar orbital altermagnet, characterized by a staggered pattern of scalar spin chiralities that is invariant under the combined operation of time reversal and fourfold rotation. We then introduce quantum spin liquids and the Schwinger boson construction, in which spins are fractionalized into bosonic spinons subject to a local constraint. Every classical phase possesses a neighboring spin liquid that inherits its discrete symmetries. The neighbor of the orbital altermagnet is an altermagnetic $\mathbb{Z}_2$ spin liquid: a phase without any spin order in which the altermagnetic chirality pattern survives alongside topological order.

1 Introduction

Altermagnets have attracted a lot of attention in the last few years [2, 3, 4]. An altermagnet is a collinear magnet whose total magnetization vanishes, but not for the reason familiar from antiferromagnets. In a conventional antiferromagnet the two spin sublattices are related by a lattice translation combined with time reversal Θ, so the moments cancel trivially. In an altermagnet the sublattices are instead related by a point-group operation combined with time reversal, for example a fourfold rotation C_{4z} on the square lattice [3]. The consequences of this symmetry structure were the subject of the preceding talks of this seminar: spin-split electronic bands despite zero net magnetization, their observation in ARPES, magnon spectra, and nontrivial band topology.

All of this physics rests on one assumption: the spins form a long-range ordered pattern, a static texture that breaks time reversal in a specific way. In this report we go to the adjacent regime. We take an altermagnetic system and make its frustrated couplings strong, so strong that quantum fluctuations prevent the spins from ever settling into a static pattern. The spin order melts, and the expectation value $\langle \hat{\boldsymbol{S}}_i \rangle$ vanishes on every site, even at zero temperature. The question is then: does anything altermagnetic survive in a phase without spin order?

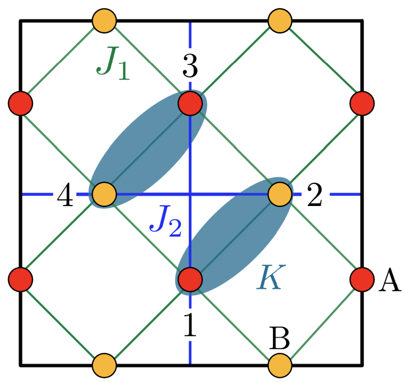

Figure 1: The checkerboard model of Eq. (1). The spins form two sublattices A and B on the bonds of a square lattice, related by the fourfold rotation C_{4z} and not by a translation. Shown are the nearest-neighbor coupling J_1, the next-nearest-neighbor coupling J_2, and the ring-exchange term K. The sites 1 to 4 label the elementary plaquette used in Eqs. (3) and (4). Adapted from Ref. [1].

Following Sobral, Mandal, and Scheurer [1], the answer is yes, but the way it survives is subtle and requires new language. The altermagnetic character does not live in the spin directions themselves but in spin-rotation-invariant observables, specifically in a staggered pattern of scalar spin chiralities. Describing the disordered phase requires the concepts of quantum spin liquids, fractionalized excitations, and emergent gauge fields, which we introduce along the way following the review by Savary and Balents [5]. The main result is a new phase of matter, the altermagnetic $\mathbb{Z}_2$ spin liquid, which combines topological order with the symmetry pattern of an altermagnet.

The question is not purely academic. Recent experiments indicate an interplay between altermagnetism and magnetic frustration in Mn_5Si_3 [6, 7] and in $CsCr_3Sb_5$ [8], so frustrated altermagnets may well exist in real materials. The report is organized as follows. Section 2 introduces a minimal spin model that is an altermagnet and can be strongly frustrated. Section 3 maps out its classical phase diagram and introduces the observables that diagnose altermagnetism without spin order. Section 4 develops the Schwinger boson description of quantum spin liquids, and Section 5 connects each classical phase to its neighboring spin liquid and identifies the altermagnetic spin liquid. Throughout we restrict ourselves to the insulating magnet; the doped, itinerant case treated in Ref. [1] is beyond the scope of this report.

2 The model

To make the question concrete we need a model that is an altermagnet and, at the same time, can be strongly frustrated. Following Ref. [1] we consider a checkerboard Heisenberg magnet with two spin sublattices A and B per unit cell, placed on the bonds of a square lattice as sketched in Fig. 1. Denoting the spin operator on bond i by $\hat{\boldsymbol{S}}_i$, the Hamiltonian reads

$$H = J_1 \sum_{\langle i,j \rangle} \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} \hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_j + K \sum_{[i,j,k,l]} \Big[(\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_j)(\hat{\boldsymbol{S}}_k \cdot \hat{\boldsymbol{S}}_l) + (\hat{\boldsymbol{S}}_i \cdot \hat{\boldsymbol{S}}_l)(\hat{\boldsymbol{S}}_k \cdot \hat{\boldsymbol{S}}_j) \Big]. \quad (1)$$

The first term is an antiferromagnetic nearest-neighbor exchange, $J_1 > 0$. The second term couples next-nearest neighbors in the checkerboard pattern and introduces frustration and competing orders. The ring-exchange term K couples any four spins on bonds sharing a common vertex in a way that preserves the C_{4v} point group, time reversal, spin rotation, and the Bravais lattice translations. Its role is to lift degeneracies and stabilize additional phases; in particular, it stabilizes the noncoplanar phase that will become central later on.

The decisive property of the model is its symmetry. With only J_1 the model is identical to a square-lattice antiferromagnet rotated by 45°; in that case an artificial translation symmetry connects the two sublattices, and the ground state would be a conventional antiferromagnet. Finite J_2 and K break this translation. What remains is the fourfold rotation C_{4z} that maps A onto B, and this is exactly the altermagnetic symmetry: the total magnetization is forced to vanish by a rotation instead of a translation. We stress that the following analysis is not limited to this particular model; it is simply a minimal explicit example in which all steps can be carried out.

3 Classical phase diagram

3.1 Classical spin ansatz

We first analyze the magnetic phases of Eq. (1) in the classical limit $S \to \infty$, where the spins become classical vectors and quantum fluctuations are absent. For the direction of the magnetization on sublattice $\mu = A, B$ in the unit cell at position $\boldsymbol{R}$ we make the general spiral ansatz

$$\boldsymbol{S}_\mu(\boldsymbol{R}) \propto \hat{\boldsymbol{a}} \cos(\boldsymbol{Q} \cdot \boldsymbol{R} - \varphi_\mu) + \hat{\boldsymbol{b}} \sin(\boldsymbol{Q} \cdot \boldsymbol{R} - \varphi_\mu) + \hat{\boldsymbol{c}}\, \eta_\mu, \tag{2}$$

where $\hat{\boldsymbol{a}}$, $\hat{\boldsymbol{b}}$, and $\hat{\boldsymbol{c}}$ are three arbitrary orthonormal vectors. The ordering wavevector $\boldsymbol{Q}$ sets the period and direction of the spiral, the phase φ_μ fixes the relative angle between the two sublattices, and η_μ is an out-of-plane canting that vanishes for all coplanar textures. Without loss of generality one sets $\varphi_A = 0$ and $\varphi_B \equiv \varphi$. For every point of the (J_2, K) plane the classical energy is then minimized over the parameters $\{Q_x, Q_y, \varphi, \eta_A, \eta_B\}$, in practice by a grid search followed by gradient descent [1]. The resulting phase diagram is shown in Fig. 2 and the corresponding spin textures in Fig. 3.

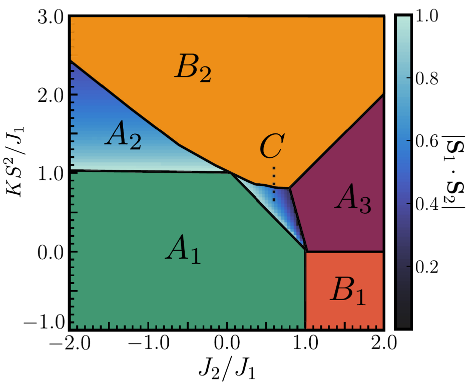

Figure 2: Classical phase diagram of the model in the (J_2, K) plane, obtained by minimizing the energy of the Ansatz (2).

3.2 Spin-rotation-invariant observables

Before walking through the phases we introduce the observables that will carry the story. The reason is the following: when we later melt the ordered phases into spin liquids, spin-rotation invariance is restored and $\langle \hat{\boldsymbol{S}}_i \rangle = 0$ on every site. Any conventional magnetic order parameter then vanishes identically. If we want to track symmetries into the disordered regime, we need quantities that are invariant under global spin rotations and can therefore remain nonzero even when there is no spin polarization.

Both observables are defined on an elementary plaquette (EP) of four spins, labeled 1 to 4 as in Fig. 1. The first is the vector of nearest-neighbor scalar products,

$$\boldsymbol{S}^{\mathrm{EP}} = (\boldsymbol{S}_1 \cdot \boldsymbol{S}_2,\ \boldsymbol{S}_2 \cdot \boldsymbol{S}_3,\ \boldsymbol{S}_3 \cdot \boldsymbol{S}_4,\ \boldsymbol{S}_4 \cdot \boldsymbol{S}_1)^T . \tag{3}$$

It is manifestly invariant under global spin rotations. If all four components are equal, the fourfold rotation C_{4z} is preserved; if they differ around the plaquette, C_{4z} is broken. $\boldsymbol{S}^{\mathrm{EP}}$ is therefore the nematicity detector in the spin-rotation-invariant sector. The second observable is the vector of scalar spin chiralities,

$$\boldsymbol{\chi}^{\mathrm{EP}} = (\boldsymbol{S}_1 \cdot (\boldsymbol{S}_2 \times \boldsymbol{S}_3),\ \boldsymbol{S}_2 \cdot (\boldsymbol{S}_3 \times \boldsymbol{S}_4),\ \boldsymbol{S}_3 \cdot (\boldsymbol{S}_4 \times \boldsymbol{S}_1),\ \boldsymbol{S}_4 \cdot (\boldsymbol{S}_1 \times \boldsymbol{S}_2))^T . \tag{4}$$

A scalar spin chirality requires three noncoplanar spins; it vanishes identically for every coplanar texture. Under time reversal all spins flip sign, and a triple product of three flipped vectors acquires a minus sign, so $\boldsymbol{\chi}^{\mathrm{EP}}$ is odd under Θ. It is the time-reversal detector in the spin-rotation-invariant sector. Together the two observables tell us whether C_{4z} and Θ are broken and, more importantly, how they are broken.

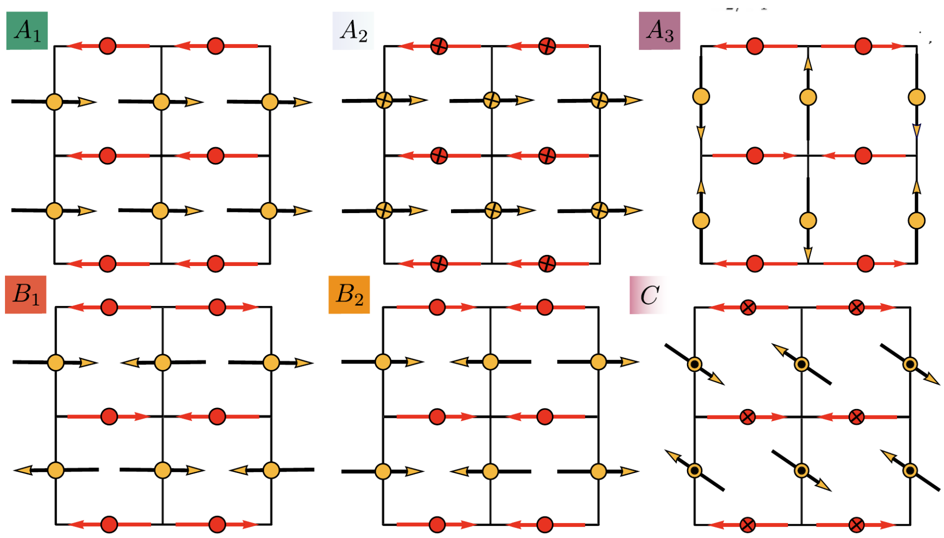

Figure 3: Classical spin textures for the six phases listed in Table 1. We can distinguish collinear, coplanar, and noncoplanar textures, as well as the presence or absence of a net magnetization. Adapted from Ref. [1].

3.3 The magnetic phases

Table 1 summarizes the six phases of the classical phase diagram together with their observables and symmetries; the last column already lists the gauge structure of the associated spin liquids that we construct in Section 5.

The anchor of the story is the collinear altermagnet A_1, stabilized at small J_2 and K. All spins on sublattice A point in one direction and all spins on B in the opposite direction. Every nearest-neighbor bond then carries the same antiferromagnetic spin product, $\boldsymbol{S}^{\mathrm{EP}} = -(1, 1, 1, 1)$, so C_{4z} is intact, and since the texture is collinear the chirality vanishes. Modulo global spin rotations, all magnetic point-group symmetries are preserved. The total

Table 1: Summary of the classical magnetic phases of Eq. (1) and their spin-rotation-invariant observables, following Ref. [1]. Here $s_0, s_0', \xi_0 > 0$ are nonuniversal numbers that depend on the couplings, M denotes the total magnetization, and the listed symmetries are those of spin-rotation-invariant observables. The last column gives the invariant gauge group (IGG) of the associated spin liquid of Section 5.

Label	Type	$\boldsymbol{S}^{\mathrm{EP}}$	$\boldsymbol{\chi}^{\mathrm{EP}}$	M	Symmetries	IGG
A_1	collinear altermagnet	$-(1,1,1,1)$	0	0	C_{4z}, σ_v, Θ	U(1)
A_2	canted altermagnet	$-(s_0,s_0,s_0,s_0)$	0	$\neq 0$	C_{4z}, σ_v, Θ	$\mathbb{Z}_2$
A_3	orthogonal antiferromagnet	$(0,0,0,0)$	0	0	C_{4z}, σ_v, Θ	$\mathbb{Z}_2$
B_1	collinear nematic	$(-1,1,-1,1)$	0	0	C_{2z}, σ_d, Θ	U(1)
B_2	collinear odd parity	$(1,1,-1,-1)$	0	0	σ_v, Θ	U(1)
C	orbital altermagnet	$-(s_0',s_0',s_0',s_0')$	$(-\xi_0,\xi_0,-\xi_0,\xi_0)$	0	$C_{4z}\Theta, \sigma_v$	$\mathbb{Z}_2$

magnetization vanishes, and it does so because of C_{4z}, not because of a translation: A_1 is a collinear altermagnet.

Increasing K leads to a second-order transition into the canted altermagnet A_2. Both sublattices develop the same out-of-plane tilt, which produces a finite magnetization $M \neq 0$. In the spin-rotation-invariant sector nothing changes: $\boldsymbol{S}^{\mathrm{EP}}$ stays uniform, only with reduced magnitude, and the chirality stays zero. At large J_2 one finds the orthogonal antiferromagnet A_3, a spiral with $\boldsymbol{Q} = (\pi, \pi)$ and $\varphi = \pi/2$ in which neighboring spins are perpendicular. All nearest-neighbor products vanish, $\boldsymbol{S}^{\mathrm{EP}} = (0,0,0,0)$, and the state can be viewed as an antiferromagnet with four spins in an enlarged unit cell. Like A_1 and A_2 it preserves all symmetries modulo spin rotations.

The phase diagram further contains two nematic phases. In B_1, reached from A_3 by a first-order transition when the sign of K is reversed, horizontal bonds carry antiparallel spins while vertical bonds carry nearly parallel spins. The spin products alternate around the plaquette, $\boldsymbol{S}^{\mathrm{EP}} = (-1,1,-1,1)$, and C_{4z} is broken down to a twofold rotation. In B_2, stabilized when $K > 0$ dominates the energetics, the components instead group pairwise, $\boldsymbol{S}^{\mathrm{EP}} = (1,1,-1,-1)$, which is a different breaking pattern of the same rotational symmetry. Both nematic phases remain coplanar, so $\boldsymbol{\chi}^{\mathrm{EP}} = 0$ and time reversal is preserved in spin-rotation-invariant observables.

The most frustrated region of the phase diagram hosts the key phase of this report: the noncoplanar phase C. Here the two sublattices cant out of the plane in opposite directions, $\eta_A = -\eta_B$. The spin products $\boldsymbol{S}^{\mathrm{EP}}$ remain uniform, so phase C is not nematic. The new feature is the chirality. Because the texture is noncoplanar, the scalar spin chiralities are nonzero, and their pattern around the plaquette is staggered,

$$\boldsymbol{\chi}^{\mathrm{EP}} \propto (-1,1,-1,1). \tag{5}$$

As shown in Fig. 4(a,b), phase C is reached from A_1 by a second-order transition at which this chirality grows continuously from zero.

The symmetry content of this pattern is the central observation. $\boldsymbol{\chi}^{\mathrm{EP}}$ is odd under time reversal alone, since every chirality flips sign. It is also odd under C_{4z} alone, since a rotation by 90° maps the pattern $(-1,1,-1,1)$ to $(1,-1,1,-1)$. The combined operation ΘC_{4z}, however, leaves the pattern invariant. This is precisely the defining symmetry structure of

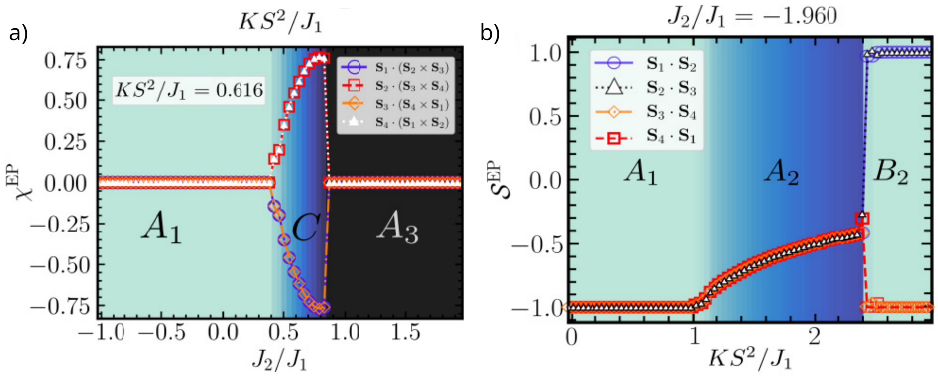

Figure 4: Scalar spin chiralities χ^{EP} (a) and nearest-neighbor spin products $\boldsymbol{S}^{\mathrm{EP}}$ (b) along one-dimensional cuts through the phase diagram; the chirality grows continuously at the second-order transition from A_1 to C. Adapted from Ref. [1].

an altermagnet, only that it now lives in the chirality channel instead of the spin channel. The name orbital altermagnet comes from the fact that in an electronic system a scalar spin chirality on a plaquette induces circulating orbital currents. The associated orbital moments are finite locally but cancel globally, again due to ΘC_{4z} rather than translation [1]. Phase C is the phase we will quantum-melt, and everything that follows is the story of what survives.

4 Quantum spin liquids and Schwinger bosons

4.1 What is a quantum spin liquid?

What do we mean when we say we quantum-melt an ordered phase into a quantum spin liquid (QSL)? A QSL is not simply a magnet without order; it is something more specific [5]. Three properties characterize it. First, there is no static magnetic order: $\langle \hat{\boldsymbol{S}}_i \rangle = 0$ on every site, even at $T = 0$. Second, the ground state is long-range entangled: it cannot be adiabatically deformed into any product state, which is what separates a QSL from a trivial paramagnet. Third, and most importantly, a QSL supports fractionalized excitations, spin-$\frac{1}{2}$ spinons that cannot be created by any single local operator, accompanied by emergent gauge fields.

A useful mental picture is Anderson's resonating valence bond (RVB) state [9]: a superposition of all possible pairings of the spins into singlets. No single pairing dominates, so the state has no spin order by construction, and one can also see that it cannot be written as a product state over any finite region. Breaking one singlet does not create a local spin flip but two unpaired spin-$\frac{1}{2}$ objects that can move apart independently; these are the spinons. We emphasize that spinons are not magnons. Magnons are the spin-1 Goldstone modes of a broken-symmetry state, whereas spinons exist precisely because no spin symmetry is broken.

The emergent gauge fields can be made more precise [5]. Any fractionalized description of a spin comes with a local constraint, and this constraint structure generates a local gauge redundancy; the QSL corresponds to a deconfined phase of the resulting emergent gauge theory. Which gauge structure emerges matters for stability. In two spatial dimensions a compact U(1) gauge theory with gapped matter is generically confining, so a U(1) spin liquid is unstable, whereas a $\mathbb{Z}_2$ gauge theory possesses a stable deconfined phase with topological order. We will meet exactly this distinction below.

4.2 The Schwinger boson representation

How do we actually construct a spin liquid? The answer used in Ref. [1] is the Schwinger boson representation [10]. Every spin operator is rewritten in terms of two bosons,

$$\hat{\boldsymbol{S}}_i = \frac{1}{2}\sum_{\sigma,\sigma'} \hat{b}^\dagger_{i\sigma}\, \boldsymbol{\sigma}_{\sigma\sigma'}\, \hat{b}_{i\sigma'}, \qquad \hat{n}_i = \sum_{\sigma} \hat{b}^\dagger_{i\sigma}\hat{b}_{i\sigma} = 2S, \tag{6}$$

where $\boldsymbol{\sigma}$ is the vector of Pauli matrices and $\sigma = \uparrow, \downarrow$ labels the boson flavor. This rewriting is exact; it embeds the spin Hilbert space into a larger bosonic one, and the local constraint $\hat{n}_i = 2S$ projects back onto the physical subspace. The price is a local U(1) gauge redundancy: the transformation $\hat{b}_{j\sigma} \to e^{i\phi_j}\hat{b}_{j\sigma}$ with arbitrary site-dependent phases ϕ_j leaves all spin operators unchanged. The bosons carry spin $\frac{1}{2}$; they are the spinons of the previous subsection.

Inserting this representation into Eq. (1) produces quartic boson interactions. These are decoupled by a mean-field Hubbard-Stratonovich transformation that introduces two complex bond fields with a transparent physical meaning. The singlet amplitude A_{ij} is the amplitude to create or annihilate a spin-0 singlet pair on the bond (i, j): one boson with spin up on site i and one with spin down on site j, combined antisymmetrically. It is Anderson's valence bond made explicit. The hopping amplitude B_{ij} is the amplitude for a boson to hop from site j to site i while preserving its spin, in analogy to tight-binding band theory. The decoupling yields the free boson Hamiltonian

$$H_b = \frac{1}{2}\sum_{i,j}\sum_{\sigma}\left(B^*_{ij}\,\hat{b}_{i\sigma}\hat{b}^\dagger_{j\sigma} - \sigma A^*_{ij}\,\hat{b}_{i\sigma}\hat{b}_{j,-\sigma} + \text{H.c.}\right) - \mu\sum_i \hat{n}_i, \tag{7}$$

with $\sigma = \pm 1$ in the second term and a Lagrange multiplier μ that enforces the constraint on average. A Fourier transformation followed by a Bogoliubov transformation diagonalizes Eq. (7) and yields the bosonic quasiparticle bands. The pattern of the parameters $\{A_{ij}, B_{ij}\}$ on the lattice, called the Ansatz, is what defines a particular spin liquid. Since a self-consistent mean-field determination of these parameters is typically not quantitatively reliable, Ref. [1] instead studies the possible Ansätze systematically; we come back to this strategy in Section 5.

4.3 Magnetic order, spin liquids, and the invariant gauge group

The connection between the ordered phases and the spin liquids is made by the bosonic band structure. The multiplier μ fixes the average boson density $\langle \hat{n}_i \rangle = 2S$, and tuning μ and the couplings changes the dispersion of the lowest band. Two situations can occur. If the lowest mode reaches zero energy, the bosons condense at some momentum, and the condensate wavefunction reconstructs exactly one of the classical spin textures of Section 3. That is magnetic order. If instead the band stays gapped, the bosons do not condense and the spins do not order. The very same Ansatz then describes a fractionalized, long-range entangled state: the neighboring spin liquid of that magnetic phase. This is the bridge between the altermagnet and the concept of a QSL.

Not all spin liquids obtained in this way are of the same type. The type is determined by the invariant gauge group (IGG), the subgroup of the U(1) gauge transformations that leaves the Ansatz invariant. If a continuous U(1) subgroup survives, the result is a U(1) spin liquid, which, as discussed in Section 4.1, is generically unstable in two dimensions. If only the sign flip $\hat{b}_i \to -\hat{b}_i$ survives, the result is a $\mathbb{Z}_2$ spin liquid, a stable topological phase. In practice, adding imaginary B_{ij} to an Ansatz generically reduces the IGG from U(1) to $\mathbb{Z}_2$, because imaginary bonds are sensitive to phase transformations that real bonds are not. The IGG therefore determines the physical nature of the spin liquid's excitations and its stability.

5 Nearby spin liquids

5.1 Matching phases to Ansätze

The strategy to find the neighboring spin liquid of each classical phase is to reverse-engineer the Ansatz [1]:

1. Choose a pattern of A_{ij} and B_{ij} on nearest-neighbor and next-nearest-neighbor bonds.
2. Diagonalize the free boson Hamiltonian (7) and find the lowest mode.
3. Tune μ such that this mode condenses.
4. Reconstruct the spin texture from the condensate and check whether it matches one of the phases of Table 1.

The minimal Ansatz whose condensation reproduces a given phase is the natural candidate, and its gapped regime is the neighboring QSL, which inherits the IGG and the discrete symmetries of the parent phase. It turns out that nearest-neighbor and next-nearest-neighbor bonds are sufficient for all six phases, without any enlargement of the unit cell.

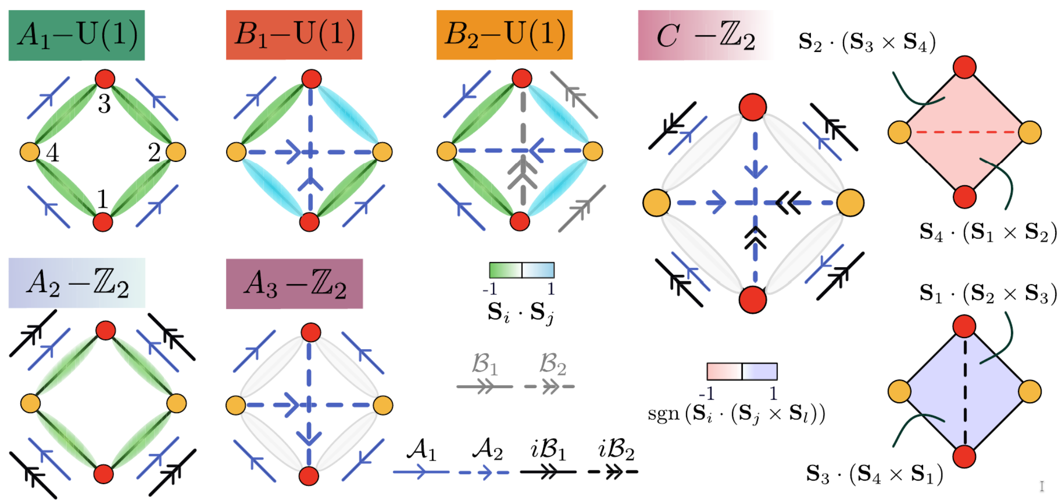

Figure 5: Schwinger boson Ansätze associated with the six magnetic phases of Table 1. Blue arrows denote real singlet amplitudes A_{ij}, black and gray arrows denote real and imaginary hopping amplitudes B_{ij}. The colored ellipses show the correlators $\langle \boldsymbol{S}^{\mathrm{EP}} \rangle$ evaluated in the gapped spin-liquid regime; for the Ansatz of phase C the staggered chirality pattern $\langle \boldsymbol{\chi}^{\mathrm{EP}} \rangle$ is nonzero as well. Adapted from Ref. [1].

Figure 5 shows the resulting Ansätze. On top of each Ansatz the correlators $\langle \boldsymbol{S}^{\mathrm{EP}} \rangle$ are drawn as colored ellipses, evaluated directly in the gapped spin-liquid regime, that is, without any condensation. One can therefore read off the spatial symmetry of each spin liquid without ever returning to the ordered phase. The result confirms the symmetry inheritance. The spin liquids of A_1, A_2, and A_3 have isotropic $\langle \boldsymbol{S}^{\mathrm{EP}} \rangle$ and are fully symmetric, while the spin liquids of B_1 and B_2 show exactly the nematic patterns of their parent phases; in all five cases $\langle \boldsymbol{\chi}^{\mathrm{EP}} \rangle = 0$. The gauge groups are listed in Table 1: the Ansätze of A_1, B_1, and B_2 give U(1) spin liquids, while those of A_2, A_3, and C involve imaginary B_{ij} or further-neighbor bonds and give $\mathbb{Z}_2$ spin liquids.

5.2 The altermagnetic spin liquid

The spin liquid associated with phase C is the main result. Its Ansatz is the only one that shows a nonzero chirality pattern in the spin-liquid regime, and its IGG is $\mathbb{Z}_2$. In the gapped regime the bosons are not condensed, so $\langle \hat{\boldsymbol{S}}_i \rangle = 0$ everywhere and spin-rotation invariance is fully restored. The correlators tell us what remains. The spin products are isotropic, so C_{4z} is intact in the spin sector. The chirality correlator, however, stays staggered, $\langle \boldsymbol{\chi}^{\mathrm{EP}} \rangle \propto (-1, 1, -1, 1)$: time reversal remains broken in spin-rotation-invariant observables, with exactly the ΘC_{4z} pattern of the orbital altermagnet.

Let us state this clearly. There is no spin order anywhere in this phase, and yet the altermagnetic arrangement of scalar chiralities survives. The phase combines $\mathbb{Z}_2$ topological order with a persistent altermagnetic chirality pattern and is therefore called the altermagnetic spin liquid [1]. It is a symmetry-enriched topological phase that was first described in this work. The altermagnetic character outlives the spin order that originally defined it.

Being a $\mathbb{Z}_2$ spin liquid, the phase also hosts vison excitations, gapped vortices of the emergent gauge field. In the altermagnetic spin liquid they are associated with local distortions of the chirality pattern, which makes them a natural target for local experimental probes [1, 11].

6 Conclusion and outlook

Three points summarize this report. First, the checkerboard J_1-J_2-K model hosts, besides collinear and nematic phases, a noncoplanar orbital altermagnet, phase C, whose staggered scalar chirality pattern is invariant under the combined operation ΘC_{4z}. Second, within the Schwinger boson construction every classical phase has a neighboring quantum spin liquid, and the discrete symmetries of the parent phase are inherited by the spin liquid, visible in the correlators $\langle \boldsymbol{S}^{\mathrm{EP}} \rangle$ and $\langle \boldsymbol{\chi}^{\mathrm{EP}} \rangle$. Third, the neighbor of the orbital altermagnet is the altermagnetic $\mathbb{Z}_2$ spin liquid: a phase without any spin order that nevertheless carries a persistent altermagnetic chirality pattern together with topological order.

On the experimental side, the most direct route to this physics would be local probes that are sensitive to bond chiralities, for example scanning tunneling microscopy or local susceptibility measurements, which could also detect the vison excitations of the $\mathbb{Z}_2$ phase [11]. The open question is whether candidate materials exist. Frustrated altermagnets are largely unexplored, but the experimental indications in Mn_5Si_3 [6, 7] and $CsCr_3Sb_5$ [8] suggest that the interplay of altermagnetism and frustration is realized in real compounds. The framework presented here also extends naturally to doped systems, where the fractionalized phases lead to spin-split Fermi surfaces without any breaking of spin-rotation symmetry; this is discussed in Ref. [1] and lies beyond the scope of this report. The paper lays the foundation for the study of altermagnetic quantum spin liquids and opens the map of frustrated altermagnets.

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